



UNIVERSITY OF SOUTHEASTERN PHILIPPINES  
*College of Teacher Education and Technology*

USeP Vehicle Reservation System

GOGOPOWERANGERS

**A PROJECT DOCUMENTATION FOR  
IT227 ADVANCE INFORMATION MANAGEMENT**

**SUBMITTED TO:  
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## **Chapter 1**

### **INTRODUCTION**

#### **A. Background of the Study**

The University of Southeastern Philippines (USEP) Tagum-Mabini Campus provides official vehicles to support academic and administrative activities such as community outreach, conferences, field research, and off-campus activities. These vehicles serve as essential resources for faculty, staff, and student organizations who require institutional transport for approved university purposes. Currently, reservations for these vehicles are processed manually through paper forms or personal inquiries. This traditional workflow often causes delays, miscommunication and difficulty in checking vehicle availability, especially the school bus which is only available at a given time and has a high demand among student organizations and academic departments [1].

Manual record-keeping systems in institutional settings are widely recognized as inefficient and prone to human error. The reliance on paper-based processes not only increases the risk of lost or misplaced records but also makes it difficult for administrators to monitor resource utilization in real time [2].

Digital reservation systems have been shown to improve scheduling accuracy, enhance user convenience, and support efficient vehicle usage monitoring [3]. When the system automates reservation requests and updates fleet status in real time, organizations reduce double booking and better optimize mobility resources [4]. The integration of policy enforcement mechanisms within systems further ensures that institutional rules, such as minimum occupancy requirements and advance booking periods, are applied consistently without manual intervention [1]. Moreover, web-based information systems have been widely adopted in university administrative operations due to their ability to centralize data, improve accessibility, and reduce the time spent on routine clerical tasks [5].

For a growing community of USeP, implementing a vehicle reservation system is a practical step toward smoother administrative operations and more effective management of university-owned vehicles. This project, the USeP Vehicle Reservation System (VRS), was developed in response to the identified gaps in the current manual process. The system aims to automate and digitize the reservation workflow while enforcing existing university policies on vehicle use.

## B. Procedures and Methods



Figure 1.1 - Data Gathering Activity

To ensure a comprehensive understanding of the existing vehicle reservation process and identify areas for improvement, the group conducted a formal interview with the Tagum-Mabini Campus Admin Office. Prior to the interview, a formal request letter was submitted and approved. The interview was designed to gather insights into the current manual system, the challenges faced in managing vehicle requests, and the specific policies governing university vehicle use. With permission, responses were recorded and later analyzed to identify the needs of the office.

## **C. Overview/Description of the Project**

### **A. The Project**

The USeP Vehicle Reservation System is designed to make reserving university vehicles easier and more organized. It allows users to submit a reservation by providing their name, destination, date of travel, and number of passengers. By automating the reservation process and integrating university policies, the system reduces manual work and ensures a smoother and more efficient way of managing vehicle requests within the university.

### **B. Scope and Limitations**

The USeP Vehicle Reservation System focuses on streamlining and organizing the process of reserving university vehicles. The system allows users to submit vehicle reservation requests by providing essential information such as their name, destination, date of travel, and number of passengers. It automates the evaluation of reservation requests by incorporating university policies, including advance booking requirements, vehicle availability, and passenger capacity considerations. Through these features, the system aims to reduce manual workload, minimize scheduling conflicts, and provide a more efficient and structured method for managing vehicle reservations within the university.

The system has a few constraints due to the current design and available resources. First, it does not include real-time vehicle tracking, meaning it cannot monitor or display the current location of any university vehicle during or after deployment. Second, the system operates locally rather than online, which limits remote access and requires all transactions to be done within the designated local environment. These limitations, while present, do not affect the system's core function of processing and managing reservation requests.

## D. Definition of Terms

- **Administrator (Admin)** - A system superuser responsible for managing user accounts, vehicle and driver records, and generating utilization reports.
- **Booking** - Synonymous with reservation in this context; refers to a confirmed or pending vehicle request in the system.
- **Driver** - A university-assigned vehicle driver who is allocated to approved reservations.
- **Occupancy Rule** - A university policy requiring that at least 50% of a vehicle's seating capacity be filled before a reservation is approved.
- **Requester** - A student or faculty member who submits a reservation request to use a university vehicle for an official or approved purpose.
- **Reservation** - The formal process of requesting a university vehicle for a specific date, time, destination, and purpose.
- **Staff** - An Admin Office employee responsible for reviewing, approving, or declining vehicle reservation requests.

## Chapter 2

### ANALYSIS OF THE CURRENT SYSTEM

#### A. Problem Cause-and-Effect Analysis

Table 2.1 – Problem Cause-and-Effect Analysis

| PIECES COMPONENT   | PROBLEM                                                 | CAUSE                                                                         | EFFECT                                                                            | PROPOSED SYSTEM                                                               |
|--------------------|---------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Performance</b> | Slow and manual processing of vehicle reservations      | Staff manually records requests using paper forms or basic logs               | Delays, long waiting time, and difficulty tracking pending reservation            | Automated reservation system that speeds up processing and schedules          |
| <b>Information</b> | Incomplete, inconsistent, or misplaced reservation data | No centralized digital record; data stored in paper-based logs                | Hard to retrieve information, inaccurate vehicle allocation, and miscommunication | Digital database storing complete and accurate reservation details            |
| <b>Economics</b>   | Wasted resources due to underutilized vehicles          | No enforced rule on minimum passenger capacity; inefficient manual assessment | Increased fuel cost and operation expenses                                        | System enforces the 50% occupancy rule and suggests the most suitable vehicle |
| <b>Control</b>     | First-come, first-served rule is difficult to monitor   | Manual listing leads to overlapping or skipped entries                        | Unfair allocation of vehicles and possible conflicts                              | System automatically logs requests in order and prevents double booking       |
| <b>Efficiency</b>  | Time-consuming approval and verification process        | Staff manually checks availability and passenger                              | Slower workflow and more staff workload                                           | Automated evaluation of policies such as availability,                        |

|                |                                                                                 |                                                       |                                                      |                                                                                                   |
|----------------|---------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                |                                                                                 | count                                                 |                                                      | capacity, and travel schedule                                                                     |
| <b>Service</b> | Requester experience inconvenience when submitting or following up reservations | Manual submission process requiring physical presence | Poor requester satisfaction and communication delays | A user-friendly system that allows users to submit reservations easily and receive updates faster |

## B. DATABASE SCHEMA(ERD)

The Entity Relationship Diagram (ERD) illustrates the database structure of the USeP Vehicle Reservation System. It shows the entities involved in the system, their attributes, and the relationships between them. The ERD ensures that data is organized efficiently and that relationships between different components of the system are clearly defined.

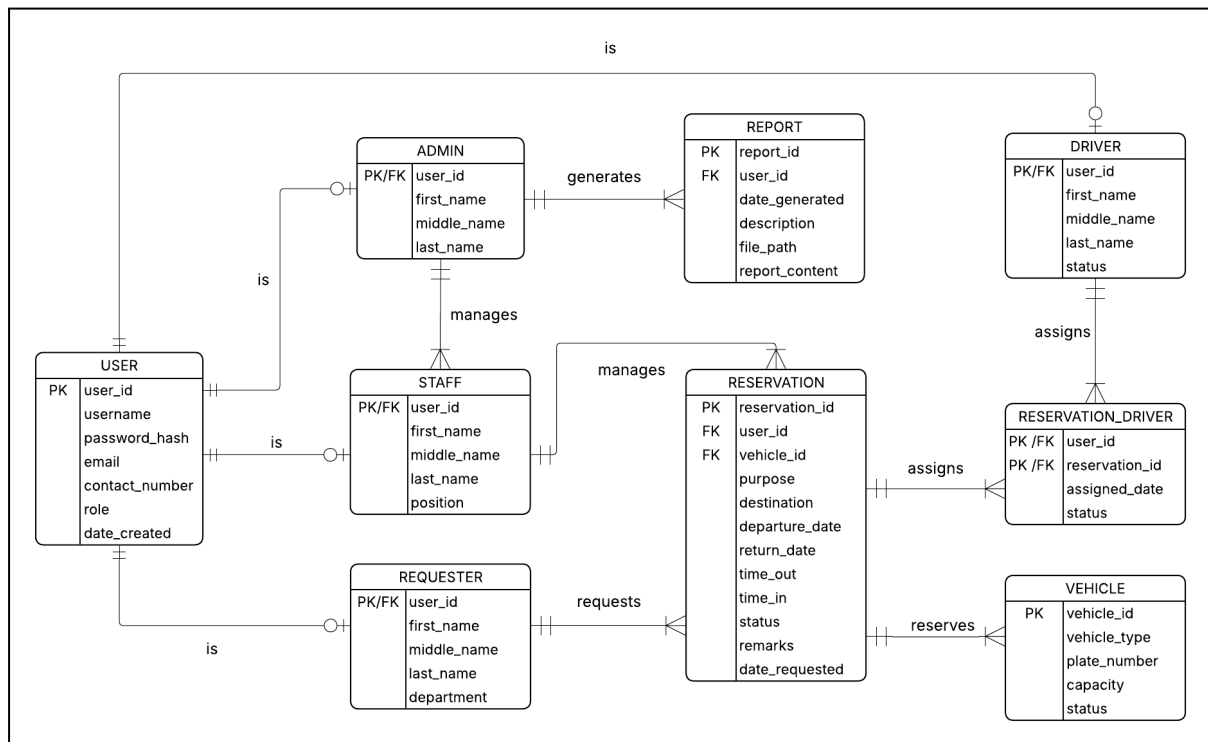


Figure 2.1 – Database Schema of USeP Vehicle Reservation System

A requester reserves a vehicle and creates a reservation. One requester can create multiple reservations, but each reservation belongs to only one requester. A reservation is requested by a requester, managed by staff, uses a vehicle, and assigns a driver through the Reservation\_Driver entity. A vehicle is reserved for reservations and is assigned to drivers, and it may serve multiple reservations over time. Drivers are assigned to reservations through the Reservation\_Driver entity, allowing one driver to handle multiple reservations. The Reservation\_Driver entity is a composite or associative entity used to resolve the many-to-many relationship between reservations and drivers. Staff members manage reservations by reviewing, approving, and monitoring reservation requests. Administrators manage staff and generate reports, while reports are generated by administrators.

## C. Table Schema

*Table 2.2 – User’s Table*

| Column Name     | Data Type    | Null | Key   | Description                                           |
|-----------------|--------------|------|-------|-------------------------------------------------------|
| user_id         | Int          | No   | PK/FK | Unique identifier for the user                        |
| username        | Varchar(50)  | No   |       | user’s username                                       |
| password_hash   | Varchar(255) | No   |       | The user's password, stored as a hash                 |
| email           | Varchar(255) | No   |       | User’s email                                          |
| contact_number  | Varchar(15)  | No   |       | User’s contact number                                 |
| role            | Varchar(50)  | No   |       | User’s role                                           |
| date_registered | Datetime     | No   |       | The date and time when the user's account was created |

*Table 2.3 – Requester’s Table*

| Column Name | Data Type    | Null | Key   | Description                                         |
|-------------|--------------|------|-------|-----------------------------------------------------|
| user_id     | Int          | No   | PK/FK | Unique identifier for the requester                 |
| first_name  | Varchar(100) | No   |       | First name of the requester                         |
| middle_name | Varchar(100) | Yes  |       | Middle name of the requester (optional)             |
| last_name   | Varchar(100) | No   |       | Last name of the requester.                         |
| department  | Varchar(100) | No   |       | The name of the department the requester belongs to |

*Table 2.4 – Staff’s Table*

| Column Name | Data Type | Null | Key   | Description                     |
|-------------|-----------|------|-------|---------------------------------|
| user_id     | Int       | No   | PK/FK | Unique identifier for the staff |

|             |              |     |  |                     |
|-------------|--------------|-----|--|---------------------|
| first_name  | Varchar(100) | No  |  | Staff's first name  |
| middle_name | Varchar(100) | Yes |  | Staff's middle name |
| last_name   | Varchar(100) | No  |  | Staff's last name   |
| position    | Varchar(50)  | No  |  | Staff's position    |

**Table 2.5 – Admin's Table**

| Column Name | Data Type    | Null | Key   | Description                     |
|-------------|--------------|------|-------|---------------------------------|
| user_id     | Int          | No   | PK/FK | Unique identifier for the admin |
| first_name  | Varchar(100) | No   |       | Admin's first name              |
| middle_name | Varchar(100) | Yes  |       | Admin's middle name             |
| last_name   | Varchar(100) | No   |       | Admin's last name               |

**Table 2.6 – Vehicle Table**

| Column Name  | Data Type   | Null | Key | Description                   |
|--------------|-------------|------|-----|-------------------------------|
| vehicle_id   | Int         | No   | PK  | Unique identifier for vehicle |
| vehicle_type | Varchar(50) | No   |     | Vehicle type                  |
| plate_number | Varchar(20) | No   |     | Vehicle plate number          |
| capacity     | Int         | No   |     | Vehicle capacity              |
| status       | Varchar(20) | No   |     | Vehicle status                |

**Table 2.7 – Driver Table**

| Column Name | Data Type    | Null | Key   | Description                      |
|-------------|--------------|------|-------|----------------------------------|
| user_id     | Int          | No   | PK/FK | Unique identifier for the driver |
| first_name  | Varchar(100) | No   |       | Driver's first name              |
| middle_name | Varchar(100) | Yes  |       | Driver's middle name             |
| last_name   | Varchar(100) | No   |       | Driver's last name               |
| status      | Varchar(20)  | No   |       | Driver's status                  |

**Table 2.8 – Reservation Table**

| Column Name    | Data Type    | Null | Key | Description                                                        |
|----------------|--------------|------|-----|--------------------------------------------------------------------|
| reservation_id | Int          | No   | PK  | Unique identifier for reservation                                  |
| user_id        | Int          | No   | FK  | The requester who made the reservation / staff who approve request |
| vehicle_id     | Int          | No   | FK  | The vehicle being reserved                                         |
| purpose        | Varchar(200) | No   |     | The purpose of the reservation                                     |
| destination    | Varchar(100) | No   |     | Destination of the trip                                            |
| departure_date | Date         | No   |     | Scheduled date of departure                                        |
| return_date    | Date         | No   |     | Expected date of return                                            |
| time_out       | Time         | No   |     | Time when the vehicle is released                                  |
| time_in        | Time         | No   |     | Time when the vehicle is reserved                                  |
| status         | Varchar(20)  | No   |     | Current reservation status                                         |
| remarks        | Varchar(100) | Yes  |     | Optional remarks about the reservation                             |
| date_requested | DateTime     | No   |     | Date when the reservation was requested                            |

**Table 2.9 – Reservation\_Driver Table**

| Column Name    | Data Type    | Null | Key   | Description                                |
|----------------|--------------|------|-------|--------------------------------------------|
| reservation_id | Int          | No   | PK/FK | Unique identifier for reservation          |
| user_id        | Int          | No   | PK/FK | Unique identifier for driver               |
| assigned_date  | DateTime     | No   |       | Date and time when the report is generated |
| status         | Varchar(200) | No   |       | Current driver status                      |

**Table 2.10 – Report Table**

| Column Name    | Data Type    | Null | Key | Description                                |
|----------------|--------------|------|-----|--------------------------------------------|
| report_id      | Int          | No   | PK  | Unique identifier for report               |
| user_id        | Int          | No   | FK  | the admin who generates report             |
| date_generated | DateTime     | No   |     | Date and time when the report is generated |
| description    | Varchar(200) | No   |     | Description of the report                  |
| file_path      | Varchar(200) | No   |     | File location where report is stored       |
| report_content | Text         | No   |     | Report content stored in text form         |

## Chapter 3

### STATEMENT OF PURPOSE OF A NEW SYSTEM

#### A. General Objectives

To develop a USeP Vehicle Reservation System that streamlines the vehicle booking process, improves organization, and enhances efficiency in managing university vehicle requests.

#### B. Specific Objectives

The specific objectives of the USeP Vehicle Reservation System, following the SMART criteria, are:

**A. To implement a vehicle reservation system** – The system will allow users to submit reservation requests by providing required details such as their name, destination, travel date, and number of passengers.

**B. To enforce university reservation policies automatically** – The system will ensure that reservations are made at least one day before travel, follow the first-come, first-served rule, and meet the minimum 50% seating capacity requirement.

**C. To automate vehicle assignment and evaluation** – The system will help staff identify the most suitable vehicle from the five available types, check availability, and approve or decline requests based on capacity and policy compliance.

**D. To maintain accurate and organized digital records** – The system will store all reservation information in a structured database, making it easier to monitor schedules, track utilization, and retrieve records when needed.

**E. To reduce manual workload for staff** – By automating the reservation process, the system will minimize manual paperwork, reduce human error, and streamline coordination between users and staff.

## **C. Potential System Benefits**

### **Tangible Benefits**

These are measurable, financial benefits that the new system provides:

- **Reduced administrative workload**

Less time spent manually recording, checking, and approving vehicle reservations, allowing staff to focus on other important tasks.

- **Lower risk of double-booking and scheduling errors**

More accurate reservation records help avoid conflicts that could lead to wasted fuel, unused trips, or corrective measures.

- **Better vehicle utilization**

The 50% occupancy rule enforced by the system ensures that vehicles are used efficiently, reducing unnecessary trips and associated fuel costs.

- **Improved Tracking and Documentation**

All reservations are stored digitally, making it easier to monitor vehicle usage, retrieve records, and generate reports. This leads to better data management and accurate monitoring of University resources.

### **Intangible Benefits**

These benefits are non-monetary but improve the university environment, brand, and user experience:

- **Improved organization and transparency**

A clear, standardized reservation process builds trust among users and staff regarding how vehicles are assigned.

- **Better communication between users and staff**

The system helps ensure that all necessary information is provided upfront, reducing misunderstandings and back-and-forth clarifications.

- **More Organized Vehicle Management**

A well-structured, automated process helps create a more orderly environment in managing transportation services, contributing to better operations within the University.

- **Positive University Image**

A modern and efficient reservation system helps demonstrate the University's commitment to improving its services. This boosts the institution's reputation for adopting updated and user-friendly technologies.

## Chapter 4

### PROPOSED SYSTEM DESIGN SPECIFICATIONS

#### A. System Architecture

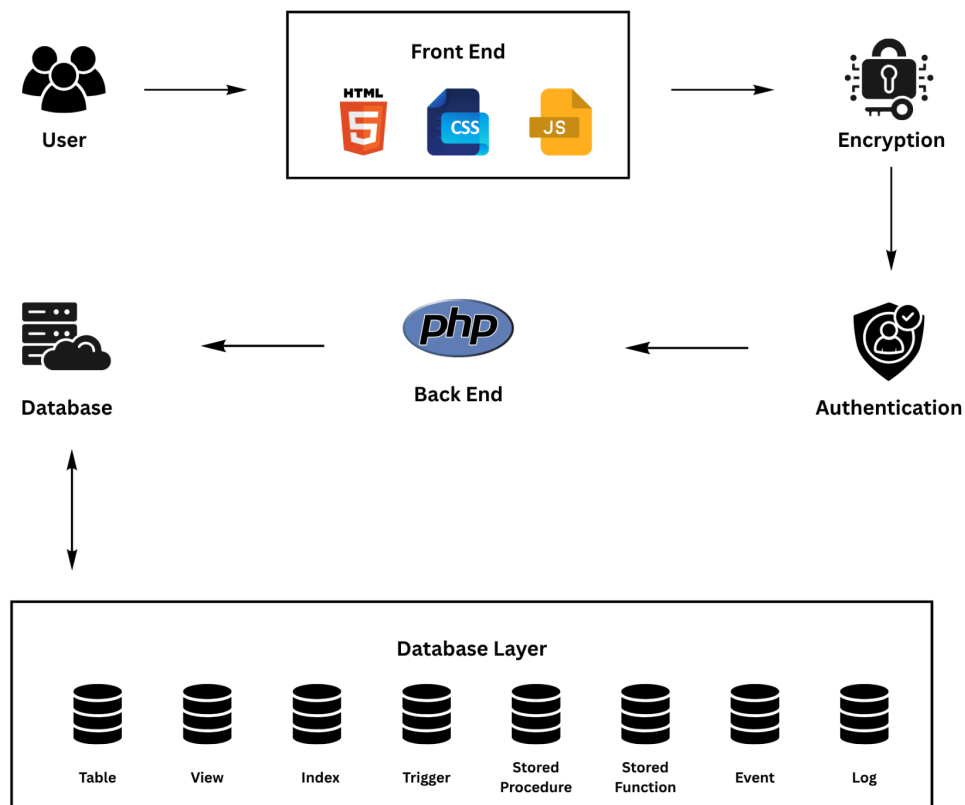


Figure 4.1 – System Architecture of USeP Vehicle Reservation System

This system architecture shows how the users, front end, back end, and database work together to process the vehicle reservations securely. User roles such as Requester can log in, submit requests, and view reservation status. The staff is responsible for reviewing vehicle reservation requests, they have the authority to approve or deny requests based on vehicle availability and schedule. The admin manages user accounts and can generate reports. The data entered by users is encrypted for security before going through authentication which then verifies user identity. Once verified, the back end handles the

processing of reservations, approving or denying requests, and updating records. The database stores the user accounts, vehicle details, and reservation data through a structured database layer.

## **B. Information Security Mechanisms**

- **Session Termination on Logout**

When the user logs out, the session associated with that user is destroyed, either by deleting session cookies or invalidating session tokens on the server side. This means the user is no longer recognized and must provide credentials again.

- **Username and Password Authentication**

The user provides a username and password. The system checks the provided credentials against a stored database to verify the identity of the user.

- **Error Handling**

The system should return generic error messages when a login attempt fails, to prevent attackers from knowing whether the username or the password is incorrect.

## **C. System Requirements (Hardware, Software, Network, Other Facilities)**

### **Hardware Requirements:**

- Minimum specifications: Intel i5 processor, 8GB RAM, 256GB SSD.
- Operating System: Windows 10 or later, or equivalent Linux distribution.

#### **Server:**

- Minimum specifications: Intel Xeon processor, 16GB RAM, 1TB HDD/SSD for data storage.
- Operating System: Windows Server 2019 or Linux - based server OS.

**Software Requirements:**

Database Management System:

- MySQL or XAMPP Server for data storage and management

Development Environment:

- Integrated Development Environment(IDE) such as Visual Studio, Eclipse, or similar for application development.

**Network Requirements:**

Local Network:

- Wired or wireless network infrastructure to support client machines within the campus.

## Chapter 5

### DOCUMENTATION

#### A. Schedule Feasibility

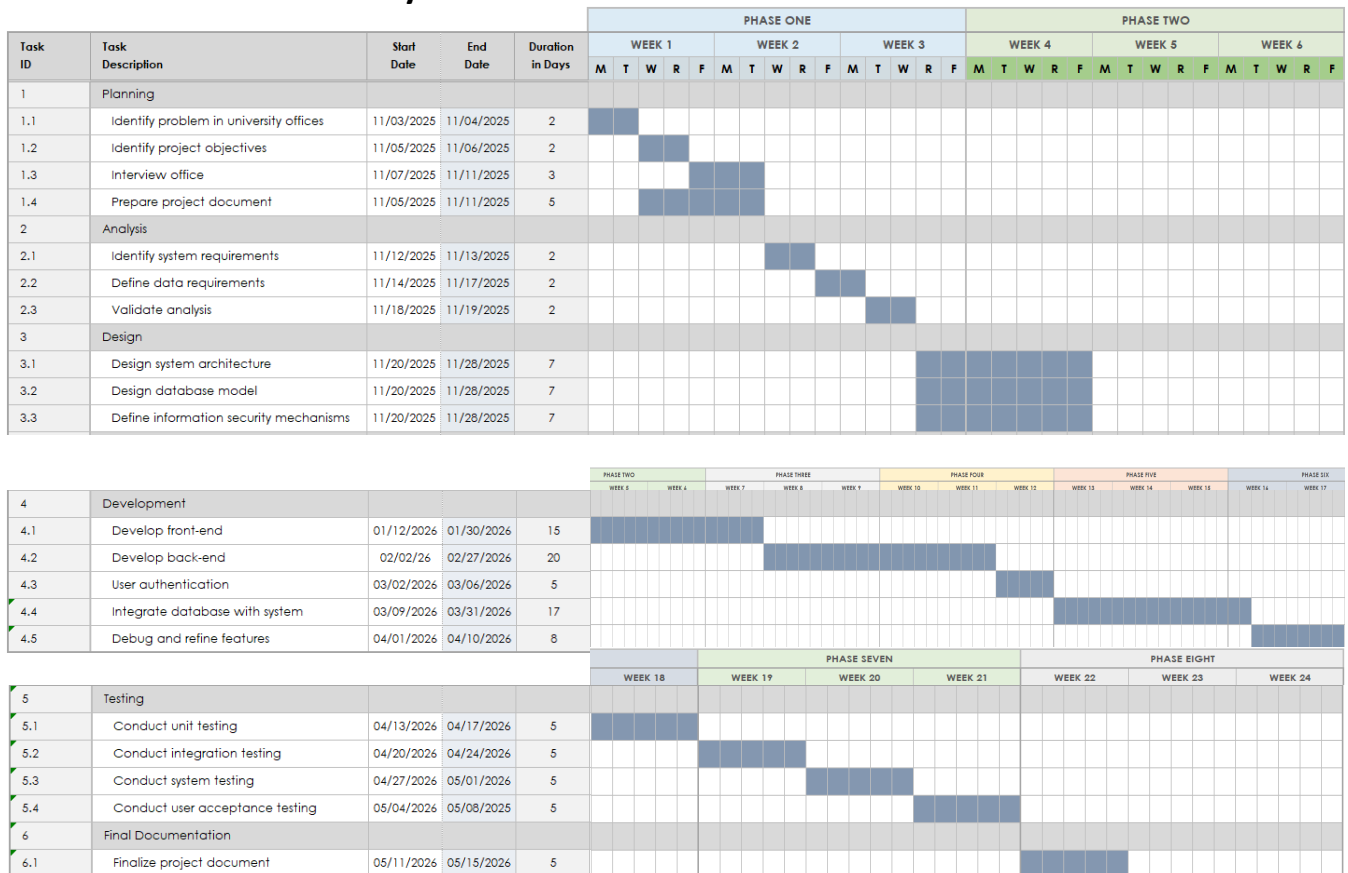


Figure 5.1 – Project's Gantt Chart

This gantt chart outlines the project timeline, showing each phase from planning to finalization. This helps monitor what needs to be done, when it should be started and completed for each task. This is used to make sure that tasks won't be delayed and each task must be done on schedule.

**B. Project Team:**



**ANONUEVO, RAUL VINCENT**

- Project Manager
- Main Programmer



**OLIVAR, BRILLIANT JOBETH B.**

- Quality Assurance (QA) Specialist
- Support Programmer



**TARIGA, SHEYNAH A.**

- System Analyst
- Support Programmer

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